

Bug Fix Impact Analysis

METIMMOX-1 Economic Evaluation

Ben Geisler

2025-11-21

Table of contents

1	Summary	1
1.1	Available Snapshot Pairs	1
2	Metadata Comparison	2
2.1	Global Parameters	2
3	Base Case Results Comparison	3
3.1	Net Monetary Benefit Comparison	3
4	PSA Results Comparison	4
4.1	Summary Statistics	4
4.2	95% Credible Intervals	4
5	Visual Comparison	5
5.1	PSA Scatter Plots (Side-by-Side)	5
5.2	PSA Scatter Plots (Overlaid)	5
6	Comparison Across Recent Issues	6
7	Summary	7

1 Summary

This report compares cost-effectiveness analysis results before and after addressing GitHub issue #70. The comparison assesses whether bug fixes or model changes affected results.

Current Comparison:

- **Issue:** #70
- **Baseline:** Commit 998dacd (2025-11-21 09:38:06)
- **Fixed:** Commit 605ff90 (2025-11-21 11:46:32)

1.1 Available Snapshot Pairs

Table 1: All Available Snapshot Pairs

	Issue	Title	Baseline Commit	Fixed Commit	Status
6	#70	PSA uses inconsistent methods for control vs. biomarker strategy predictions	998dacd	605ff90	Current
5	#68	[P4] Include Weibull model for reference	4665f93	998dacd	
1	#42	[P0] Critical Bug: Incorrect death transition calculation in calculate_outcomes.R	607b539	cadf81a	
4	#67	[P4] Move snapshot folder	f3674e8	d5d1356	
2	#62	[P4] Move cached file	d046c7c	f3674e8	
3	#64	[P3] impact of bug fixes	8681d1e	8681d1e	

2 Metadata Comparison

Table 2: Metadata Comparison

Item	Baseline	Fixed
Git Commit	998dacd	605ff90
Timestamp	2025-11-21 09:38:06	2025-11-21 11:46:32
R Version	R version 4.5.1 (2025-06-13 ucrt)	R version 4.5.1 (2025-06-13 ucrt)
Issue Number	70	70

2.1 Global Parameters

Table 3: Global Parameter Comparison

	Parameter	Baseline	Fixed	Changed
WTP	WTP	51,000	51,000	No
time_horizon	time_horizon	520	520	No
n_samples	n_samples	5,000	5,000	No
n_sim	n_sim	5,000	5,000	No
cl	cl	0.01923077	0.01923077	No
dr	dr	0.04	0.04	No
DSA_mult	DSA_mult	0.2	0.2	No
USE_BOTH_MODELS	USE_BOTH_MODELS	0	0	No

3 Base Case Results Comparison

Table 4: Base Case Cost-Effectiveness Results Comparison

Strategy	Cost (€)				Effect (QALYs)				ICER (€/QALY)		
	Baseline	Fixed	Difference	% Change	Baseline	Fixed	Difference	% Change	Baseline	Fixed	Difference
control	21,748.34	20,705.58	-1,042.77	-4.79	1.8040	1.8444	0.0403	2.23	—	—	—
crp	55,992.46	54,636.38	-1,356.08	-2.42	1.9153	1.7845	-0.1308	-6.83	307,907	—	—
tlr	76,439.41	84,729.83	8,290.42	10.85	1.2957	1.7038	0.4081	31.50	—	—	—
tmb_braf	68,032.43	62,245.99	-5,786.44	-8.51	1.8235	1.8696	0.0461	2.53	—	1,645,887	—

3.1 Net Monetary Benefit Comparison

Table 5: Net Monetary Benefit Comparison at WTP = €51,000

	Strategy	Baseline (€)	Fixed (€)	Difference (€)
1	control	70,257.69	73,356.74	3,099.06
2	crp	41,685.57	36,372.76	-5,312.81
4	tmb_braf	24,963.72	33,103.52	8,139.80
3	tlr	-10,359.72	2,163.83	12,523.55

Optimal Strategy: Unchanged (control)

4 PSA Results Comparison

4.1 Summary Statistics

Table 6: PSA Mean Costs and Effects (n = 5,000 simulations)

Strategy	Mean Cost (€)			Mean Effect (QALYs)		
	Baseline	Fixed	Difference	Baseline	Fixed	Difference
control	20,936.34	20,747.36	-188.98	1.6999	1.8435	0.1436
crp	54,118.74	54,856.22	737.48	1.6977	1.8837	0.1859
tlr	85,905.64	86,892.87	987.23	1.6230	1.7947	0.1717
tmb_braf	64,950.69	65,757.95	807.25	1.7942	1.9835	0.1893

4.2 95% Credible Intervals

Table 7: PSA 95% Credible Intervals

Strategy	Cost Baseline		Cost Fixed		Effect Baseline		Effect Fixed	
	2.5%	97.5%	2.5%	97.5%	2.5%	97.5%	2.5%	97.5%
control	15,783.79	27,129.21	15,676.15	26,817.10	1.1101	2.3569	1.3388	2.0390
crp	40,595.71	69,484.76	16,137.87	152,119.33	1.1276	2.2533	0.9958	3.4041
tlr	61,363.13	115,121.66	15,140.97	163,776.53	1.0532	2.1950	0.5740	3.2621
tmb_braf	47,458.10	85,506.50	17,309.20	155,120.39	1.1067	2.5413	1.0243	3.3431

5 Visual Comparison

5.1 PSA Scatter Plots (Side-by-Side)



Figure 1: Cost-effectiveness plane showing PSA simulations baseline (left) and fixed (right)

5.2 PSA Scatter Plots (Overlaid)

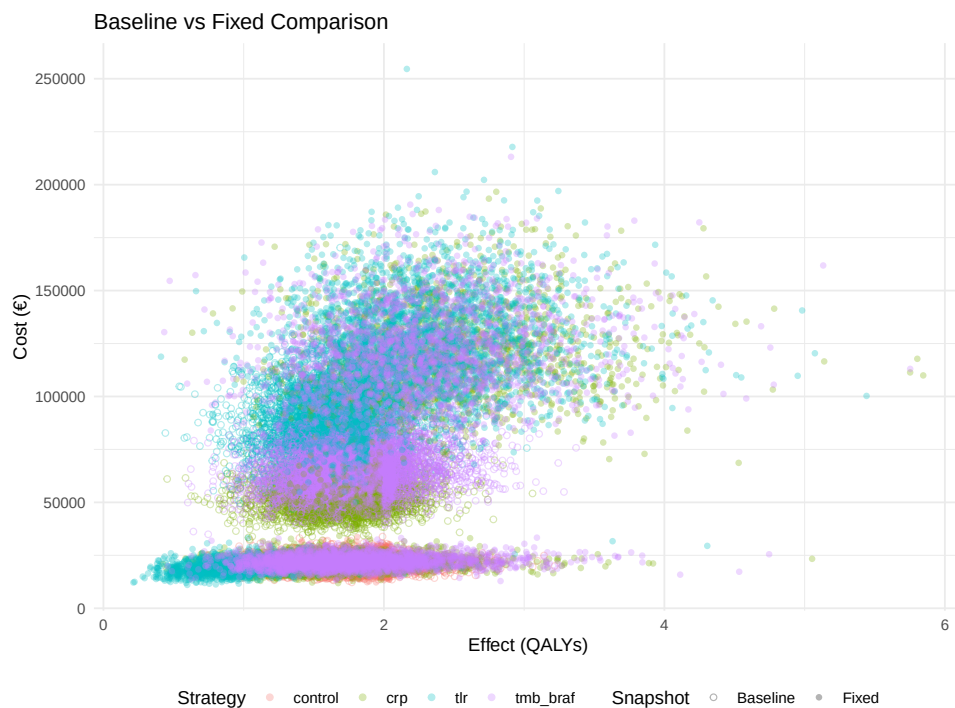


Figure 2: Cost-effectiveness plane with overlaid baseline and fixed PSA simulations

6 Comparison Across Recent Issues

This table summarizes the impact of bug fixes across the 5 most recent issues:

Table 8: Summary of Bug Fix Impacts (5 Most Recent Issues)

Issue	Title	Optimal (Baseline)	Optimal (Fixed)	Changed?	Max Cost Change (€)	Max Effect Change (QALYs)	Max ICER Change (€/QALY)
#70	PSA uses inconsistent methods for control vs. biomarker strategy predictions	control	control	No	8,290.42	0.4081	-Inf
#68	[P4] Include Weibull model for reference	control	control	No	0.00	0.0000	0
#42	[P0] Critical Bug: Incorrect death transition calculation in calculate_outcomes.R	control	control	No	0.00	0.0000	0
#67	[P4] Move snapshot folder	control	control	No	0.00	0.0000	0
#62	[P4] Move cached file	control	control	No	0.00	0.0000	0

7 Summary

The comparison for issue #70 shows:

- **Optimal Strategy:** Unchanged (control)
- **Maximum Cost Change:** €8,290.42 (substantial)
- **Maximum Effect Change:** 0.408 QALYs (substantial)
- **Maximum ICER Change:** €-Inf/QALY (minor)
- **Parameter Changes:** None detected

Interpretation: Changes exceed typical thresholds and should be carefully reviewed to ensure they align with the intended bug fix.

Report completed on: 2025-11-21

Repository: ben-geisler/METIMMOX-1

Report version: 1.1